

ECO101

Introduction to Microeconomics

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Credit: Cengage 2014

Week 6 – Welfare & Efficiency

- *What is consumer surplus? How is it related to the demand curve?*
- *What is producer surplus? How is it related to the supply curve?*
- *What are price ceilings and price floors? How do they affect market outcomes?*
- *How do taxes affect market outcomes? How do the effects depend on whether the tax is imposed on buyers or sellers?*
- *What is the incidence of a tax? What determines the incidence?*

Consumer Surplus (CS)

Consumer surplus is the amount a buyer is willing to pay minus the amount the buyer actually pays:

$$CS = \text{Max Price} - \textit{Price}$$

<i>name</i>	<i>max Price</i>
Mike	\$300
Chester	250
Brad	150
Joe	125

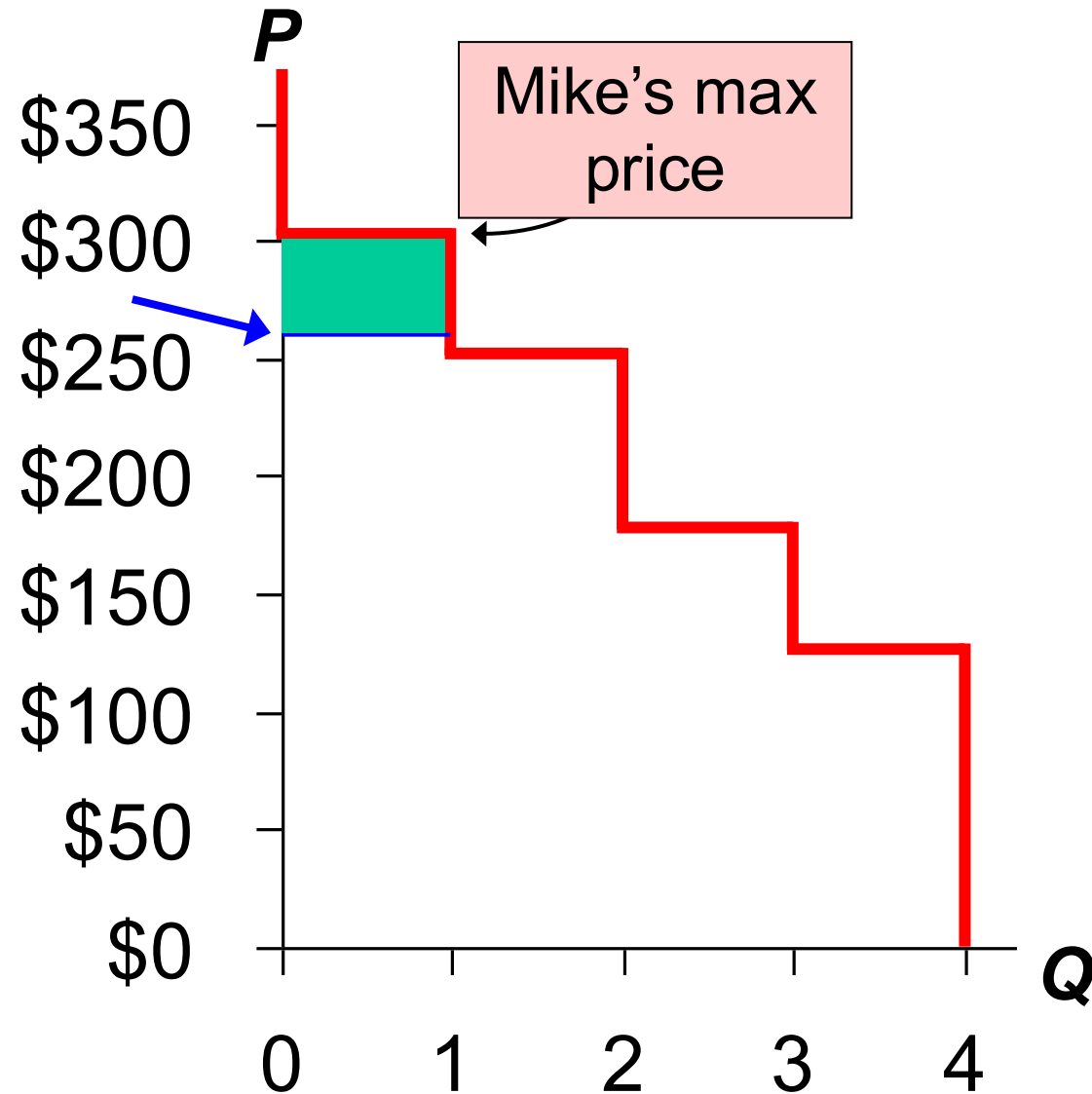
Suppose ***P*** = \$260.

Mike's CS = \$300 – 260 = \$40.

The others get no CS because they do not buy an iPod at this price.

Total CS = \$40.

CS and the Demand Curve

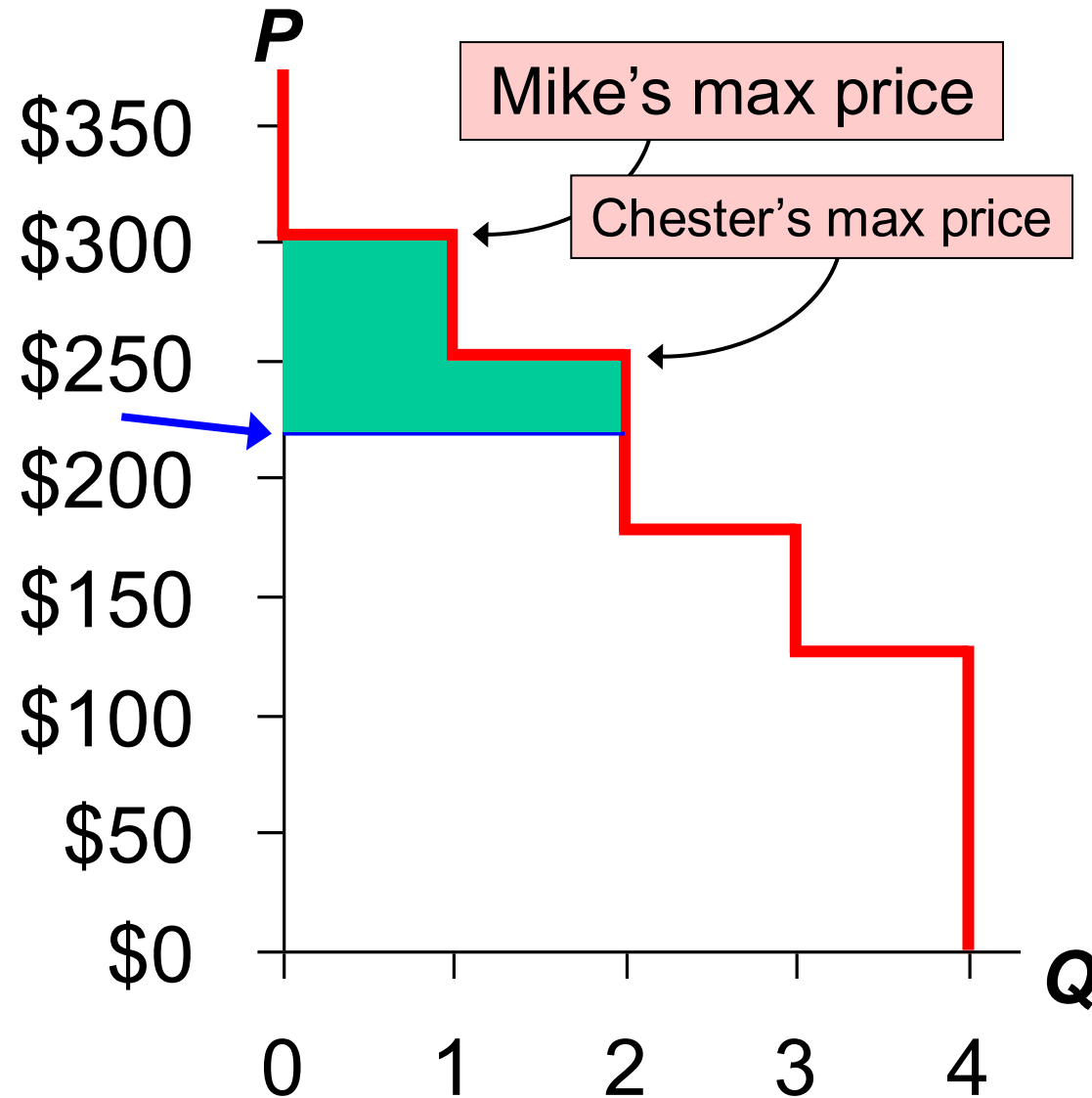


$$P = \$260$$

$$\text{Mike's CS} = \$300 - 260 = \underline{\$40}$$

$$\text{Total CS} = \underline{\$40}$$

CS and the Demand Curve



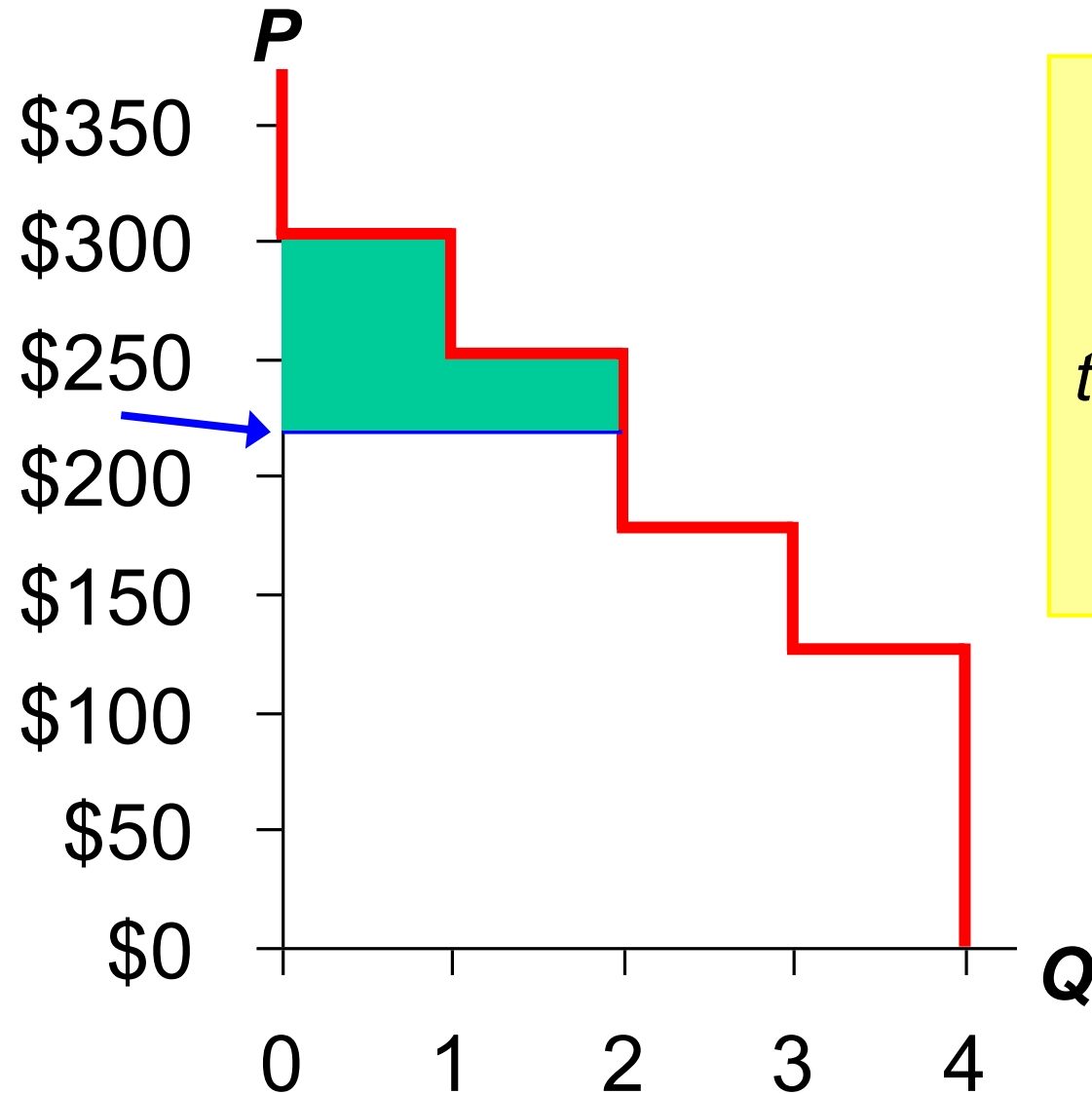
Instead, suppose
 $P = \$220$

Mike's CS =
 $\$300 - 220 = \underline{\$80}$

Chester's CS =
 $\$250 - 220 = \underline{\$30}$

Total CS = \$110

CS and the Demand Curve



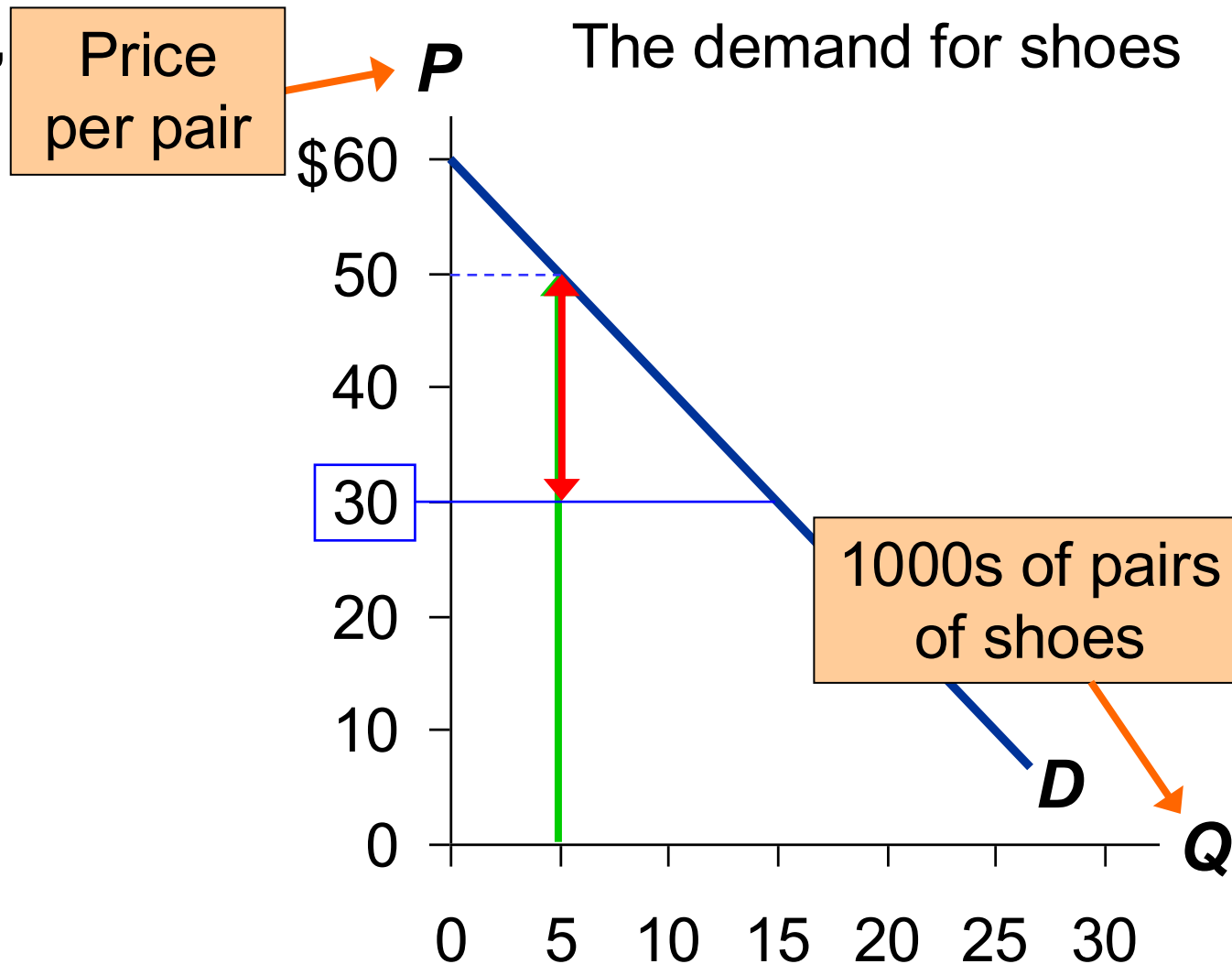
*The lesson:
Total CS equals
the area under
the demand curve
above the price,
from 0 to Q .*

CS with Lots of Buyers & a Smooth D Curve

At $Q = 5$ (thousand), the marginal buyer is willing to pay \$50 for pair of shoes.

Suppose $P = \$30$.

Then his consumer surplus = \$20.



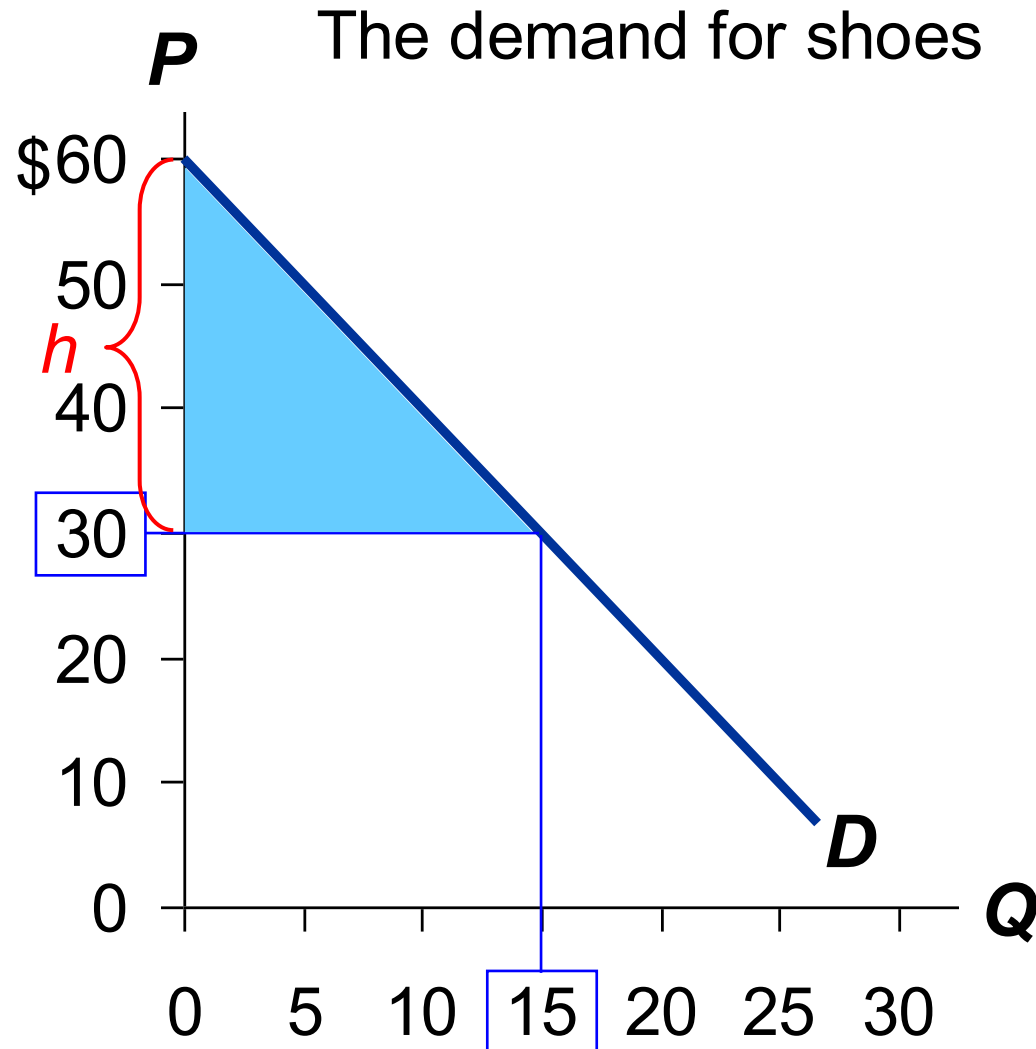
CS with Lots of Buyers & a Smooth D Curve

CS is the area b/w
P and the **D** curve,
from 0 to **Q**.

Recall: area of
a triangle equals
 $\frac{1}{2} \times \text{base} \times \text{height}$

Height =
 $\$60 - 30 = \underline{\$30}$.

So,
 $\text{CS} = \frac{1}{2} \times 15 \times \30
 $= \underline{\$225}$.



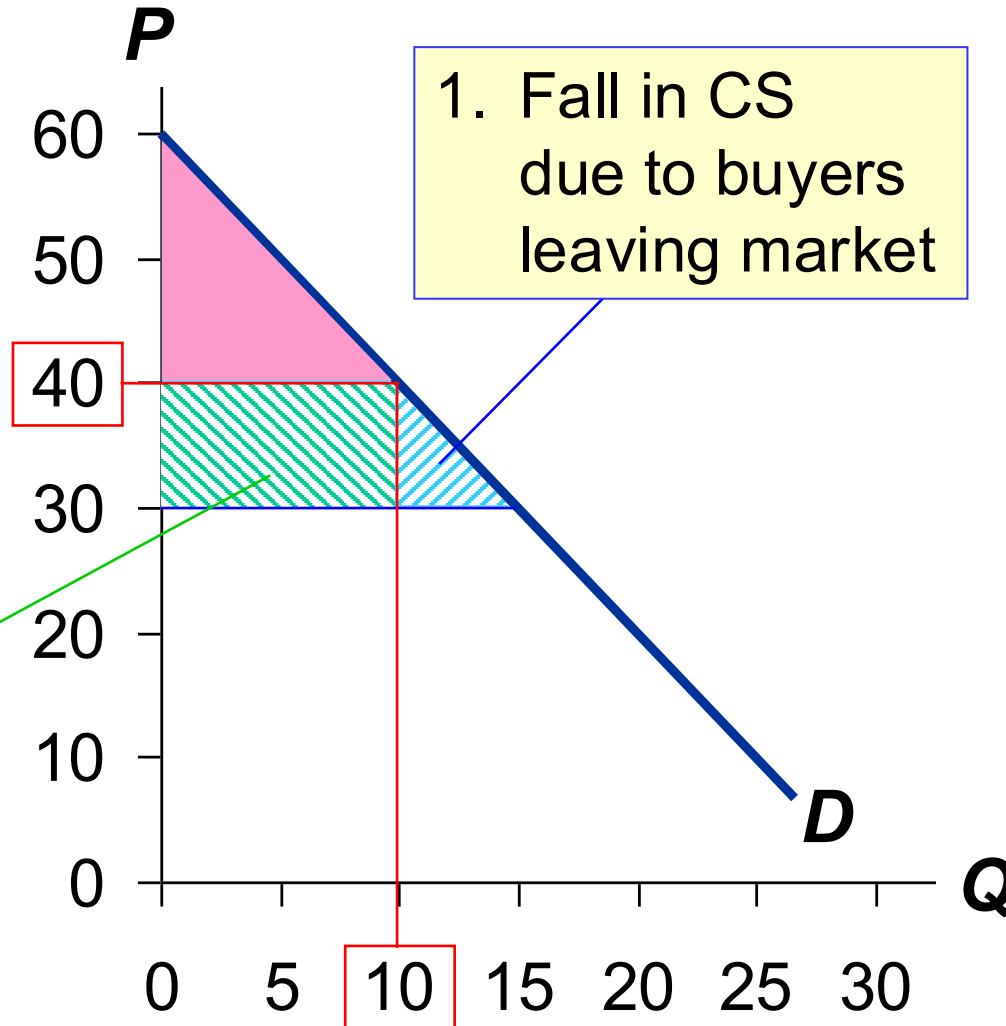
How a Higher Price Reduces CS

If P rises to \$40,

$$\begin{aligned} CS &= \frac{1}{2} \times 10 \times \$20 \\ &= \$100. \end{aligned}$$

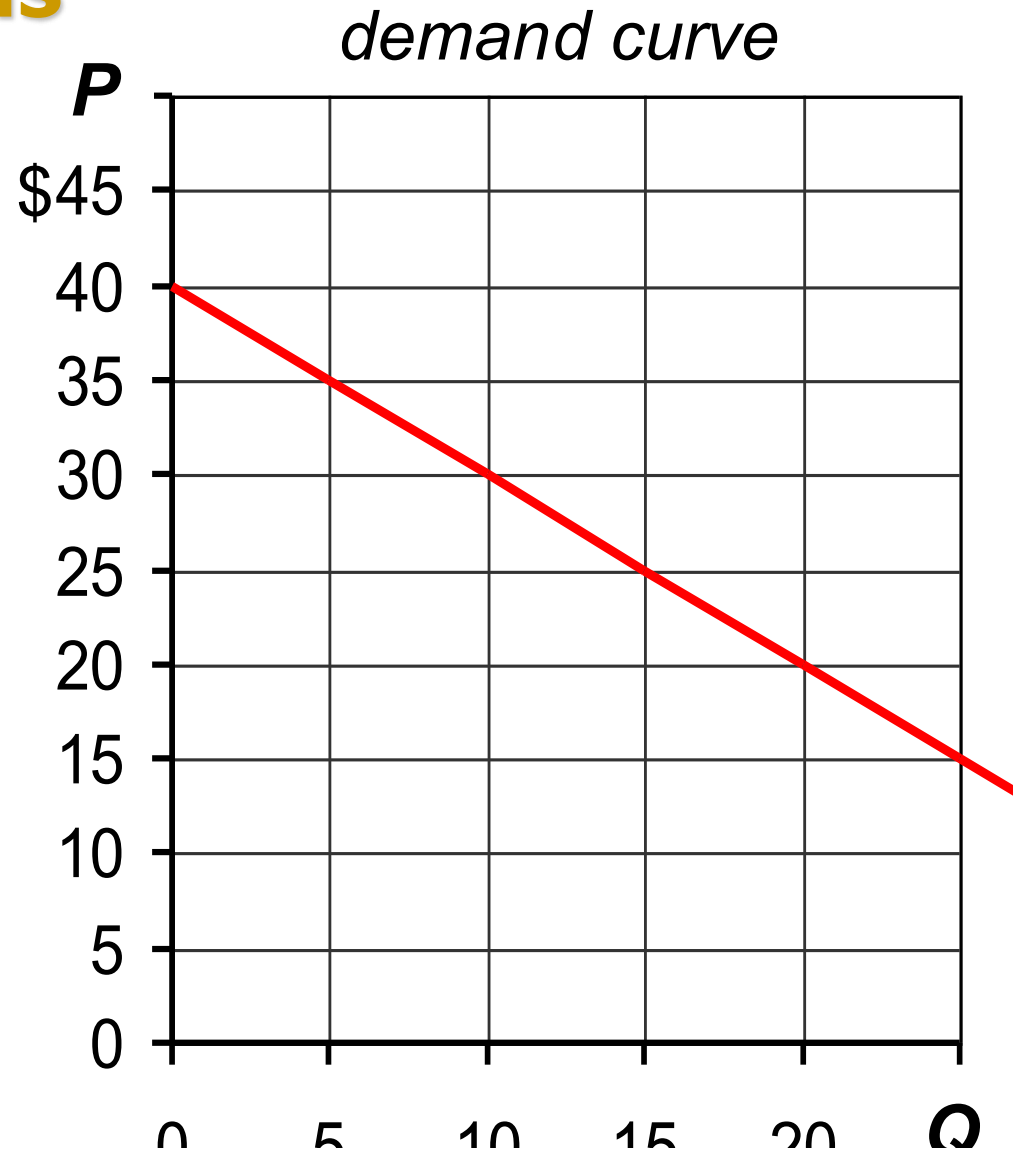
Two reasons for the fall in CS.

2. Fall in CS due to remaining buyers paying higher P



Consumer Surplus

- A. Find marginal buyer's max price at $Q = 10$.
- B. Find CS for $P = \$30$.
Suppose P falls to \$20.
How much will CS increase due to...
- C. buyers entering the market
- D. existing buyers paying lower price



Consumer Surplus

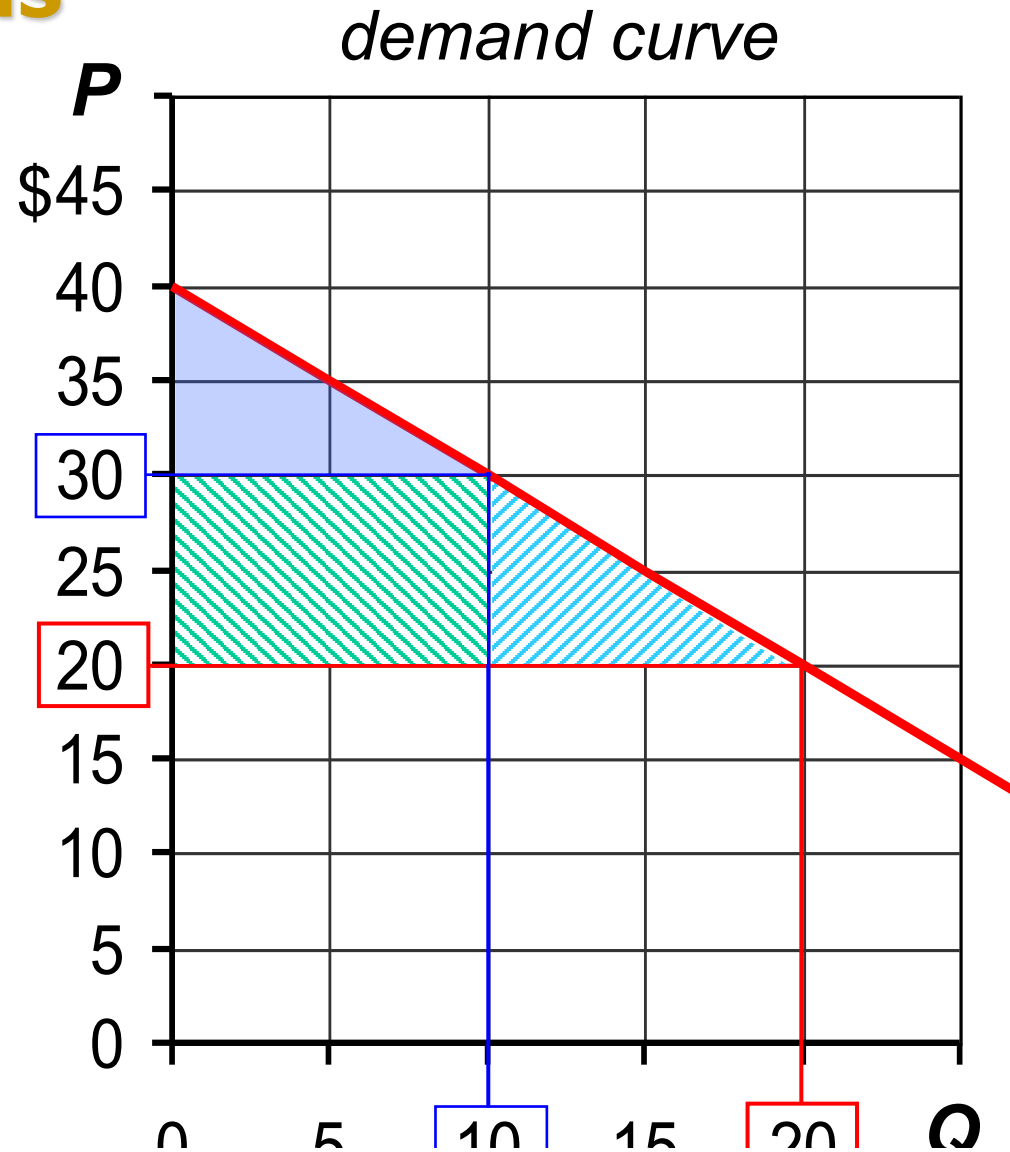
A. At $Q = 10$, marginal buyer's max price is \$30.

B. $CS = \frac{1}{2} \times 10 \times \10
 $= \underline{\$50}$

P falls to \$20.

C. CS for the additional buyers
 $= \frac{1}{2} \times 10 \times \$10 = \underline{\$50}$

D. Increase in CS
on initial 10 units
 $= 10 \times \$10 = \underline{\$100}$



Cost and the Supply Curve

- **Cost** is the value of everything a seller must give up to produce a good (i.e., opportunity cost).
- Includes cost of all resources used to produce good, including value of the seller's time.
- Example: Costs of 3 sellers in the lawn-cutting business.

<i>name</i>	<i>cost</i>
Billy	\$10
Tre	20
Mike	35

A seller will produce and sell the good/service only if the price exceeds his or her cost.

Hence, cost is a measure of willingness to sell.

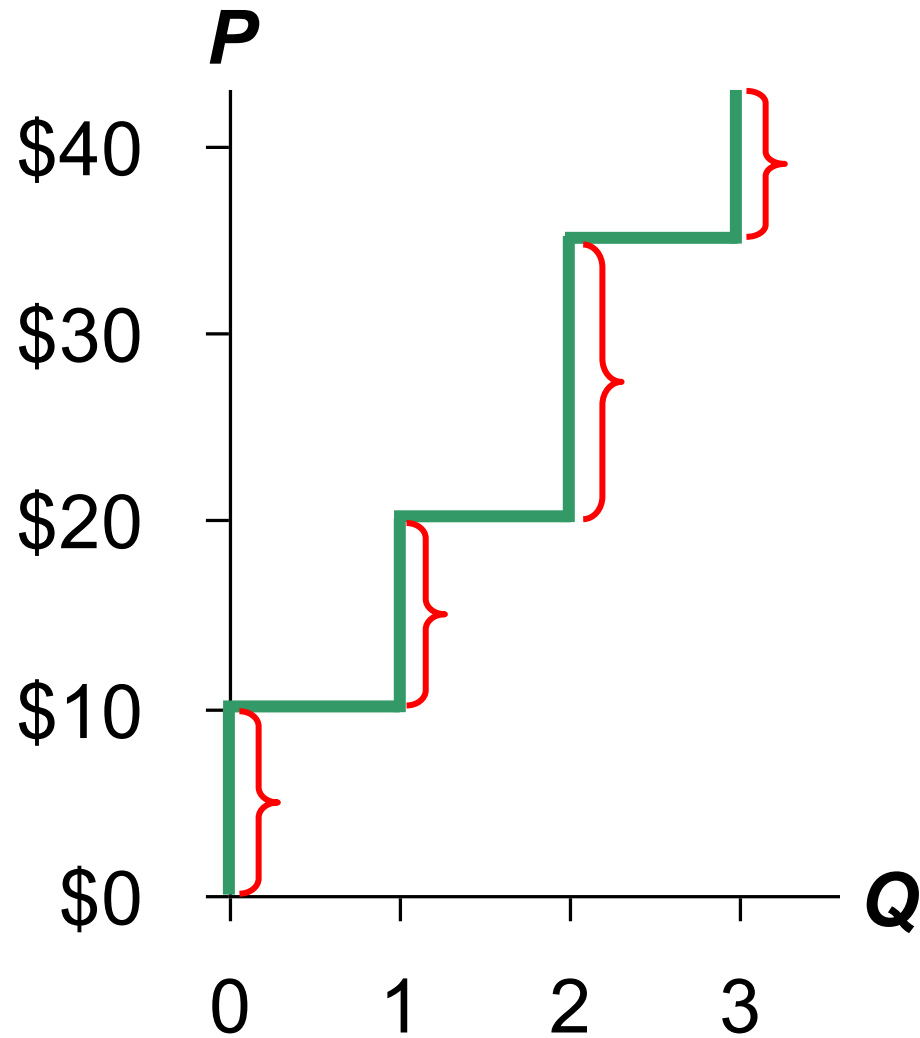
Cost and the Supply Curve

Derive the supply schedule
from the cost data:

<i>name</i>	<i>cost</i>
Billy	\$10
Tre	20
Mike	35

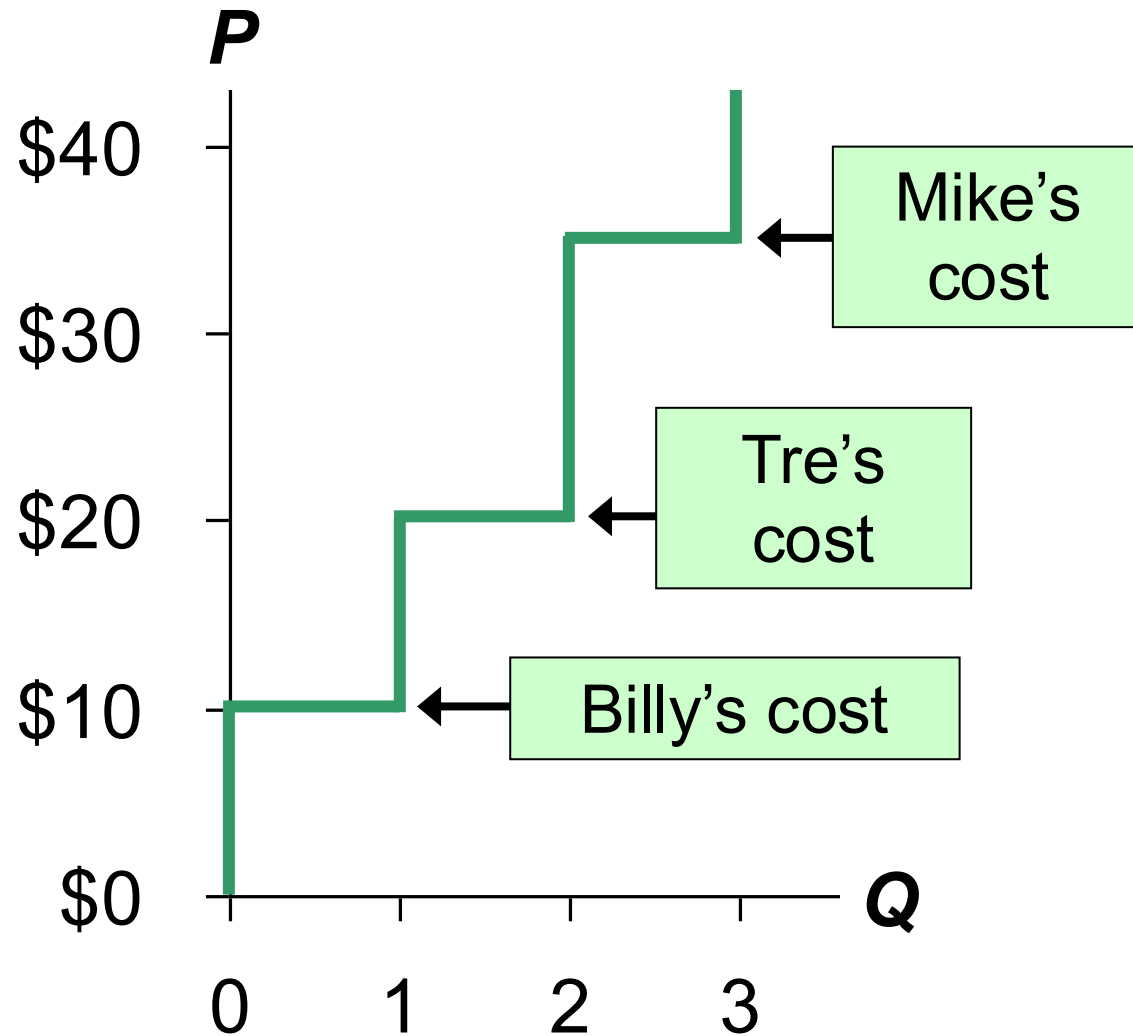
<i>P</i>	<i>Q^s</i>
\$0 – 9	0
10 – 19	1
20 – 34	2
35 & up	3

Cost and the Supply Curve



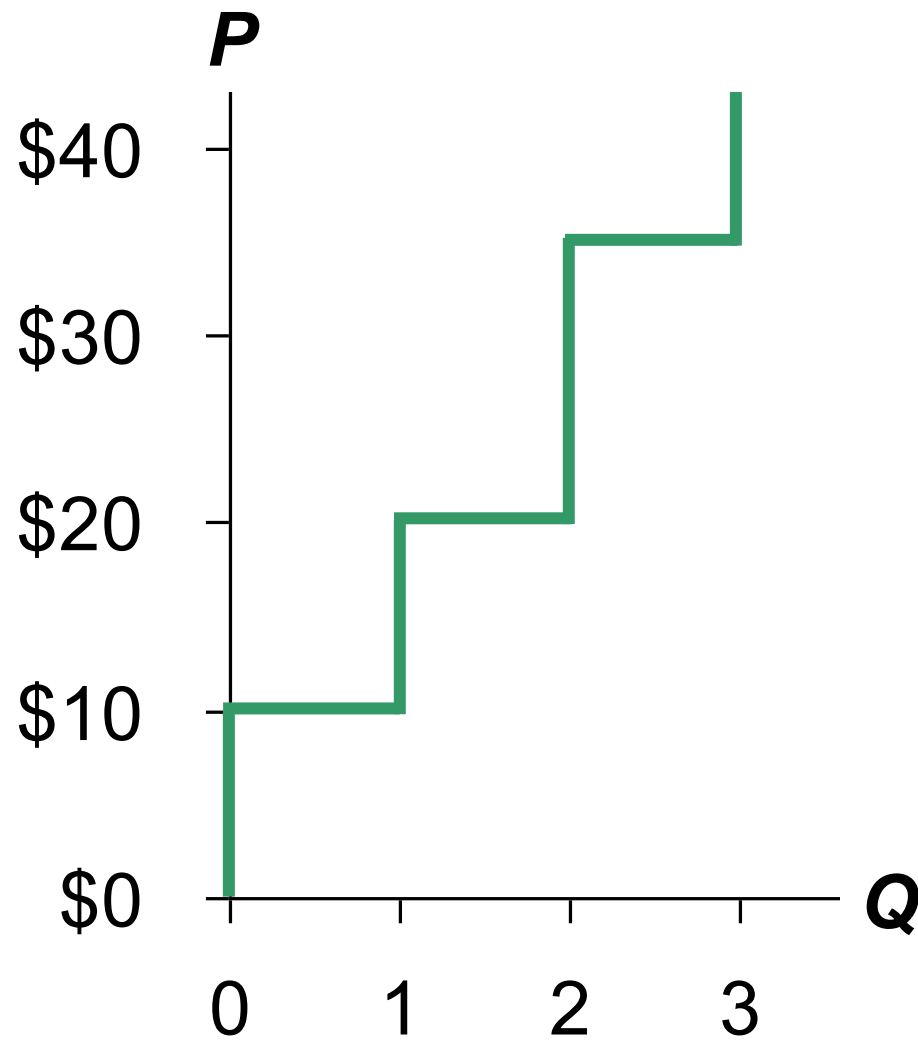
P	Q^s
\$0 – 9	0
10 – 19	1
20 – 34	2
35 & up	3

Cost and the Supply Curve



At each Q , the height of the S curve is the cost of the *marginal seller*, the seller who would leave the market if the price were any lower.

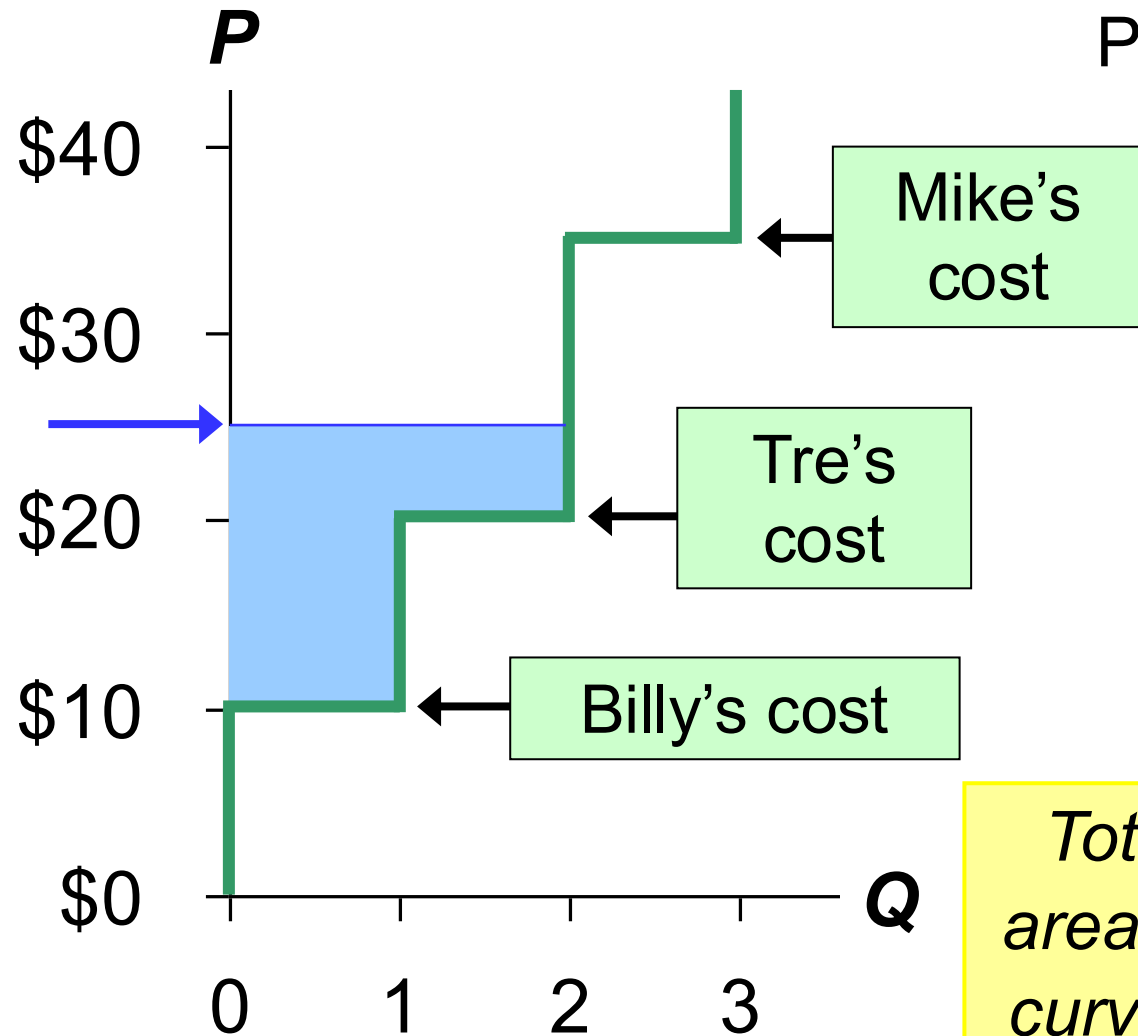
Producer Surplus



$$PS = P - \text{cost}$$

Producer surplus (PS):
the amount a seller
is paid for a good
minus the seller's cost

Producer Surplus and the S Curve



$$PS = P - \text{cost}$$

Suppose $P = \$25$.

Billy's PS = \$15

Tre's PS = \$5

Mike's PS = \$0

Total PS = \$20

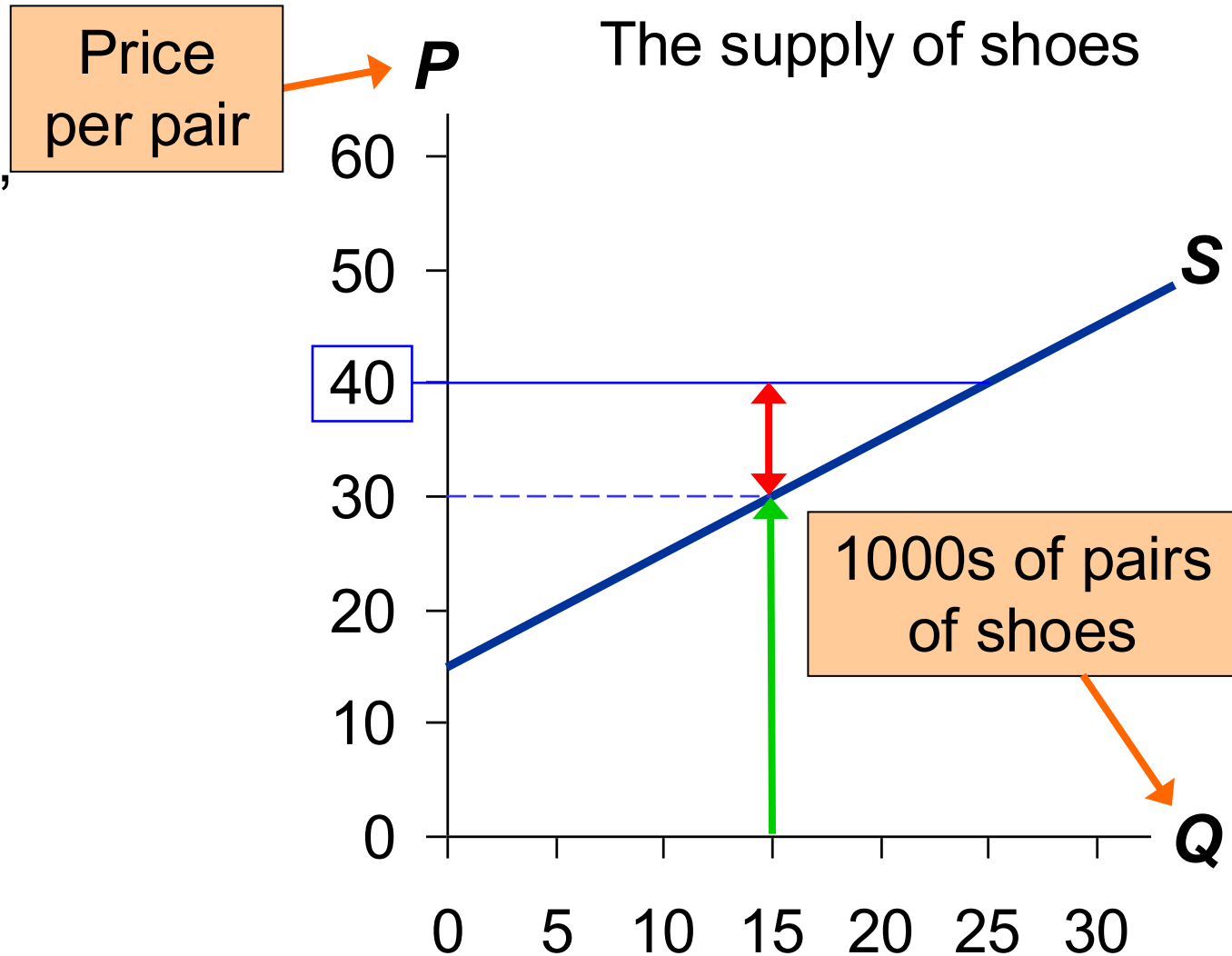
Total PS equals the area above the supply curve under the price, from 0 to Q .

PS with Lots of Sellers & a Smooth S Curve

Suppose $P = \$40$.

At $Q = 15$ (thousand),
the marginal seller's
cost is \$30,

and her producer
surplus is \$10.

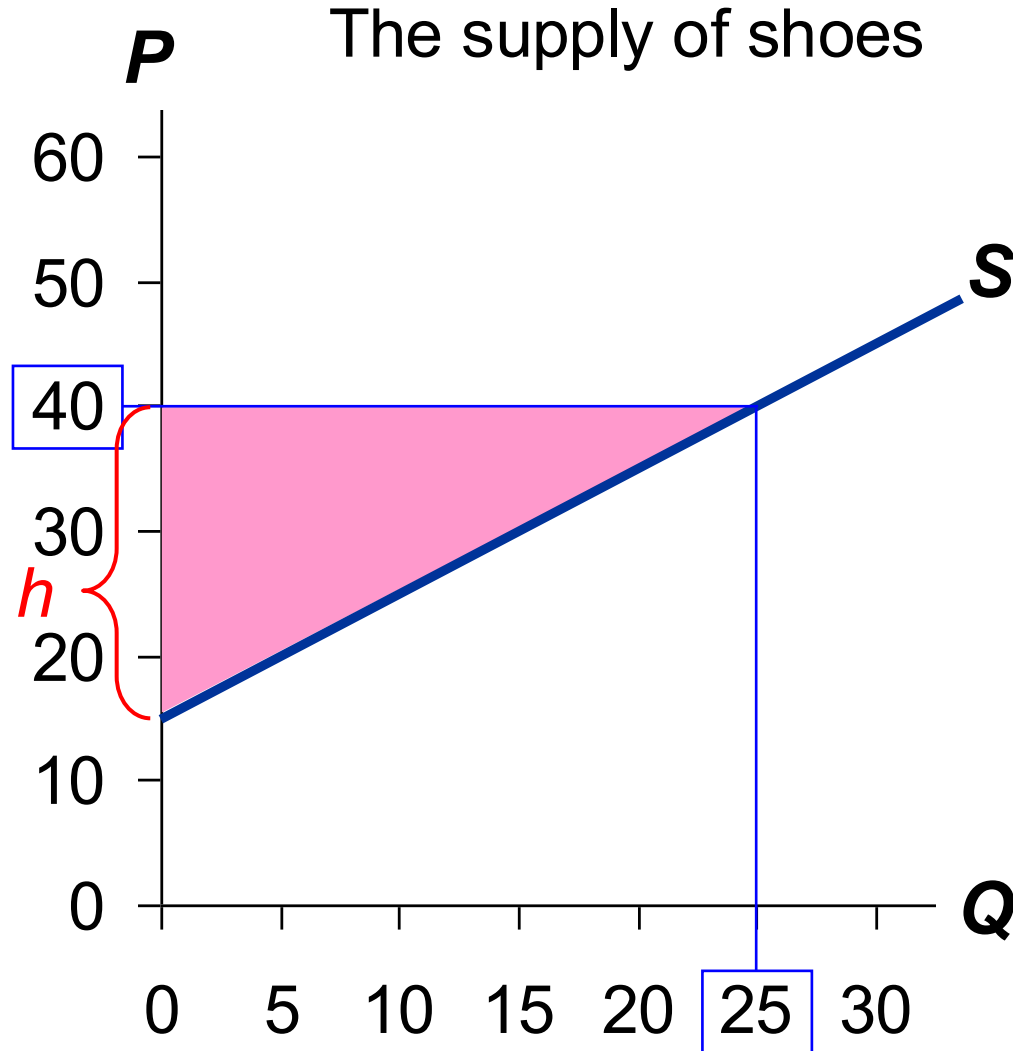


PS with Lots of Sellers & a Smooth S Curve

PS is the area b/w
P and the **S** curve,
from 0 to **Q**.

The height of this
triangle is
 $\$40 - 15 = \25 .

So,
 $PS = \frac{1}{2} \times b \times h$
 $= \frac{1}{2} \times 25 \times \25
 $= \underline{\underline{\$312.50}}$



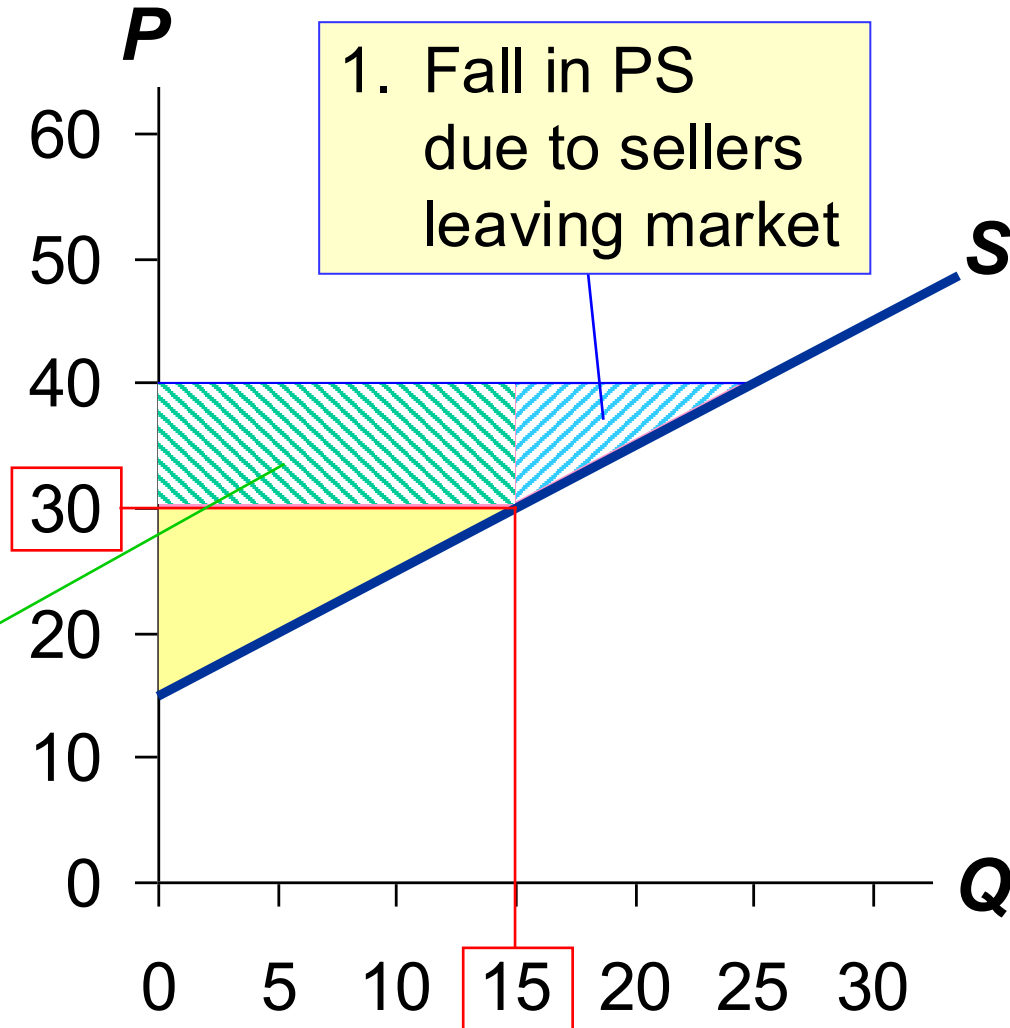
How a Lower Price Reduces PS

If P falls to \$30,

$$\begin{aligned} PS &= \frac{1}{2} \times 15 \times \$15 \\ &= \underline{\underline{\$112.50}} \end{aligned}$$

Two reasons for the fall in PS.

2. Fall in PS due to remaining sellers getting lower P



1. Fall in PS due to sellers leaving market

Producer Surplus

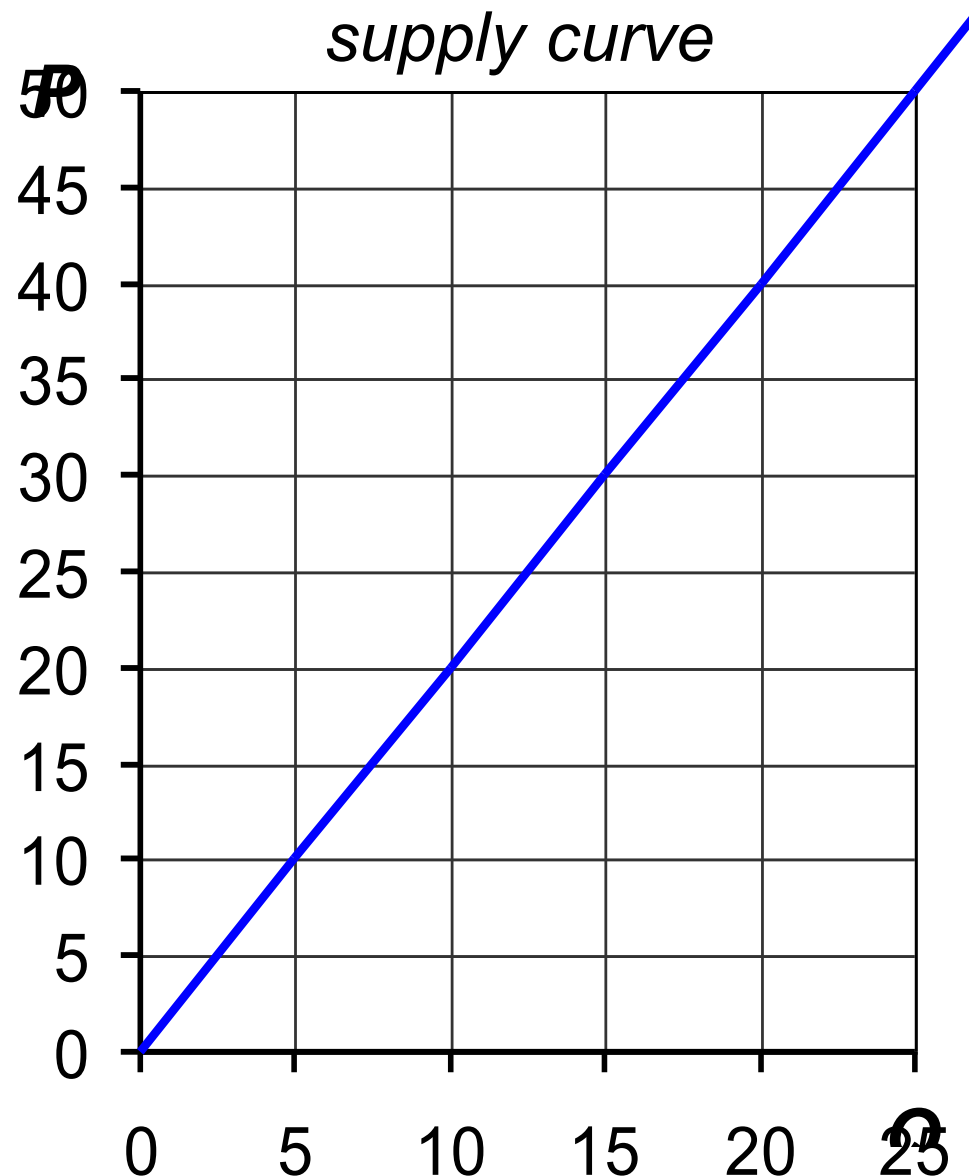
A. Find marginal seller's cost at $Q = 10$.

B. Find total PS for $P = \$20$.

Suppose P rises to \$30.
Find the increase in PS due to:

C. selling 5 additional units

D. getting a higher price on the initial 10 units



Producer Surplus

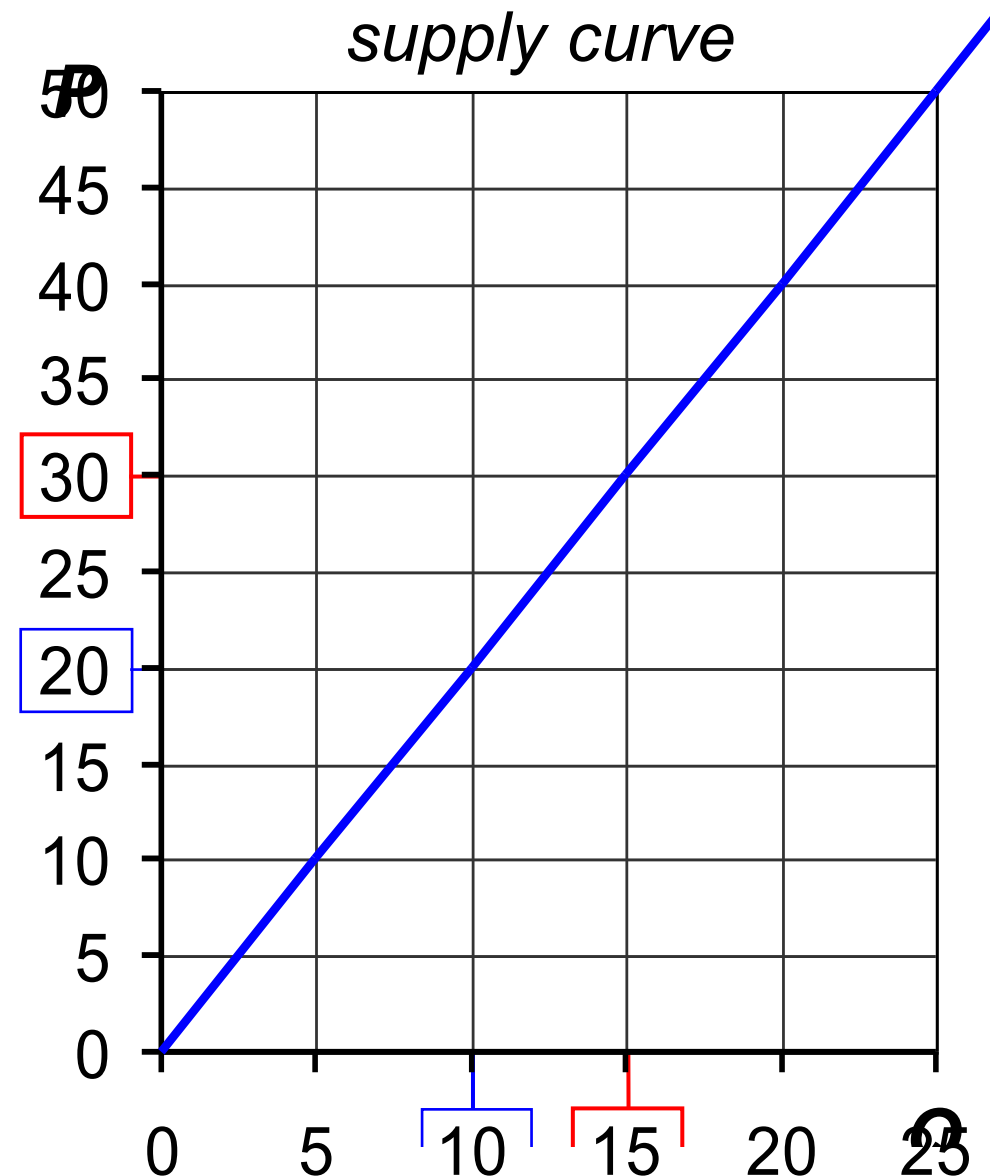
A. At $Q = 10$,
marginal cost = \$20

B. $PS = \frac{1}{2} \times 10 \times \20
= \$100

P rises to \$30.

C. PS on
additional units
= $\frac{1}{2} \times 5 \times \10 = \$25

D. Increase in PS
on initial 10 units
= $10 \times \$10$ = \$100



CS, PS, and Total Surplus

CS = (max value to buyers) – (amount paid by buyers)
= buyers' gains from participating in the market

PS = (amount received by sellers) – (cost to sellers)
= sellers' gains from participating in the market

Total surplus = CS + PS
= total gains from trade in a market
= (value to buyers) – (cost to sellers)

Government Interventions that Alter the Private Market Outcome

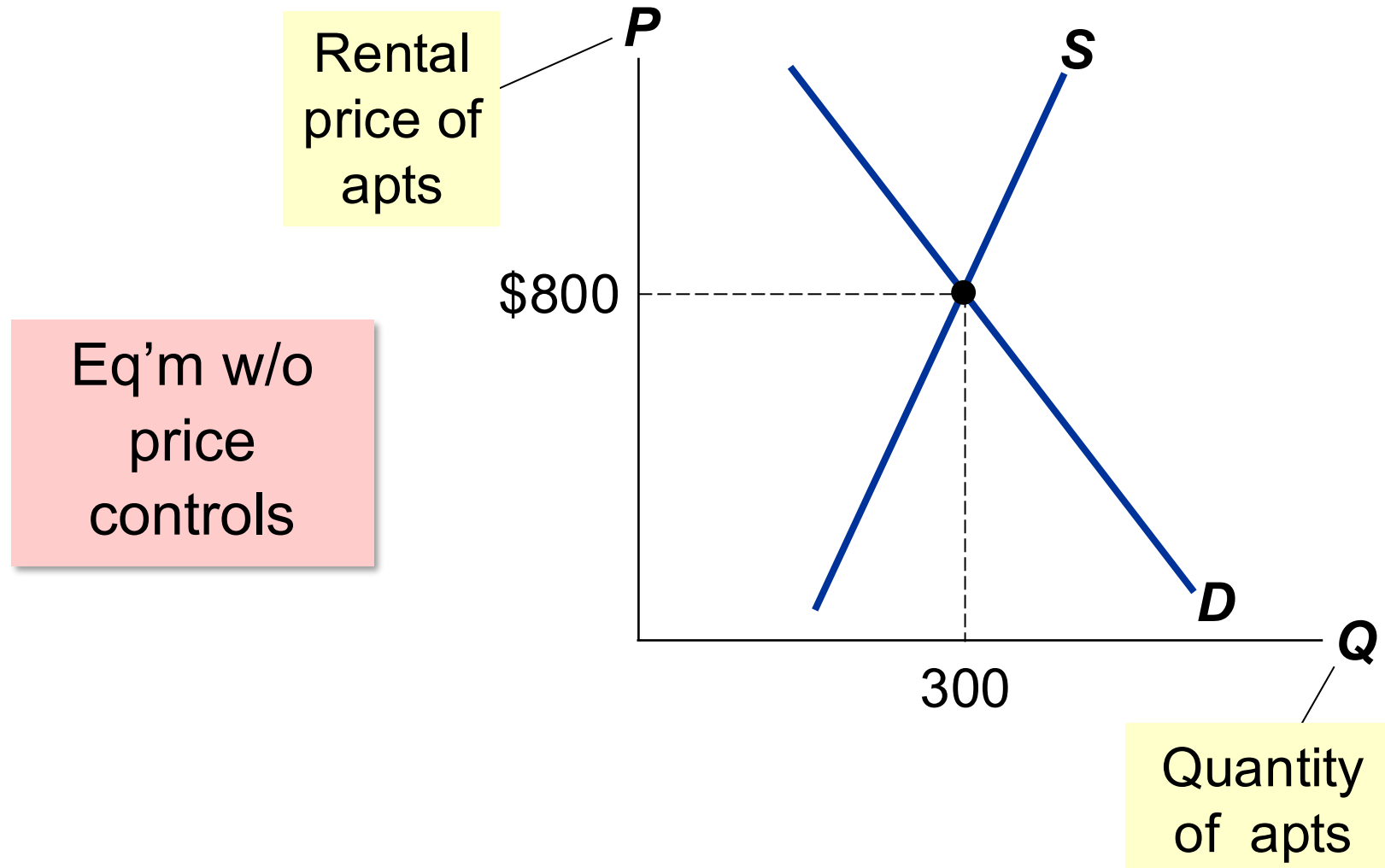
Price controls

Price ceiling: a legal maximum on the price of a good or service *Example: rent control*

Price floor: a legal minimum on the price of a good or service *Example: minimum wage*

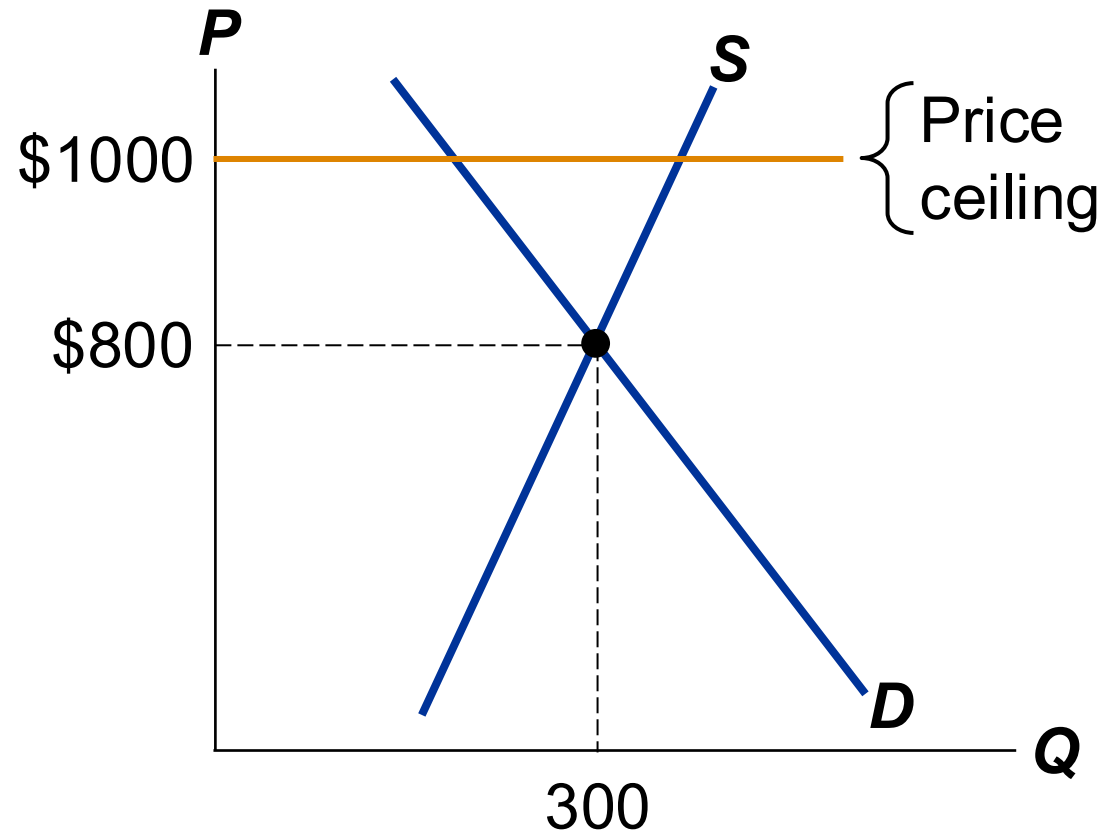
Taxes: The govt can make buyers or sellers pay a specific amount on each unit.

EXAMPLE 1: The Market for Apartments



How Price Ceilings Affect Market Outcomes

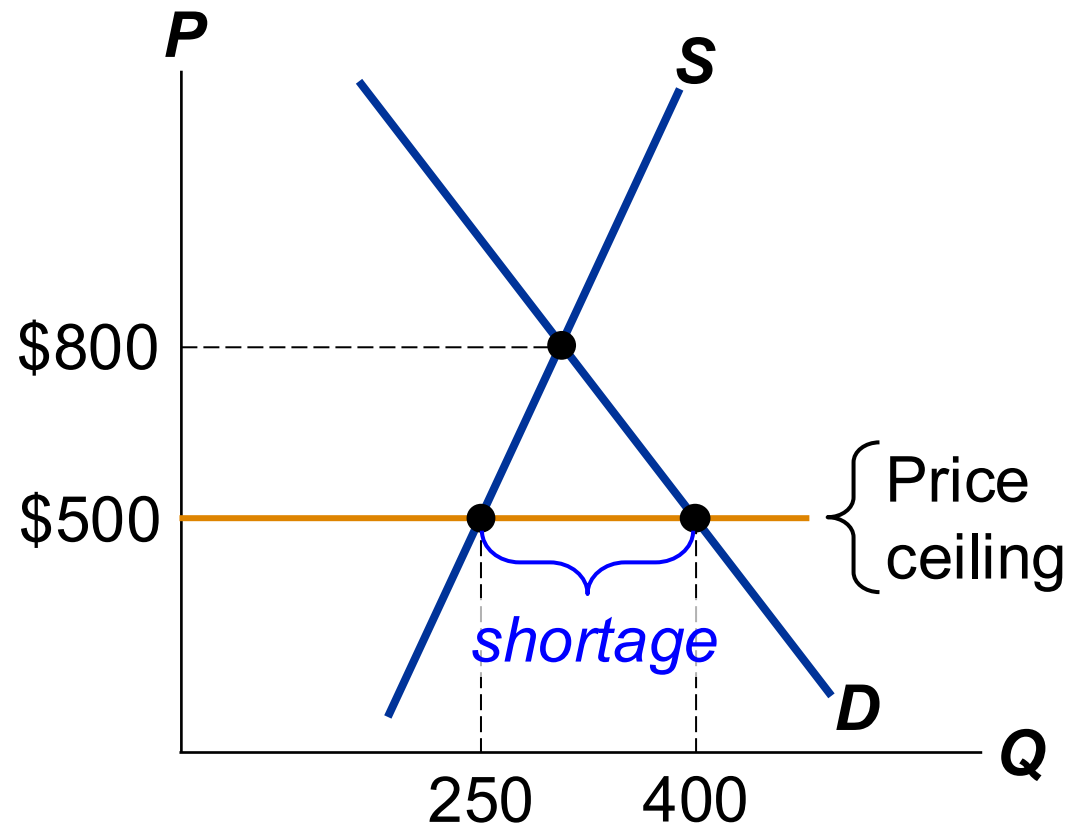
A price ceiling above the eq'm price is **not binding**—has no effect on the market outcome.



How Price Ceilings Affect Market Outcomes

The eq'm price (\$800) is above the ceiling and therefore illegal.

The ceiling is a **binding constraint** on the price, causes a shortage.



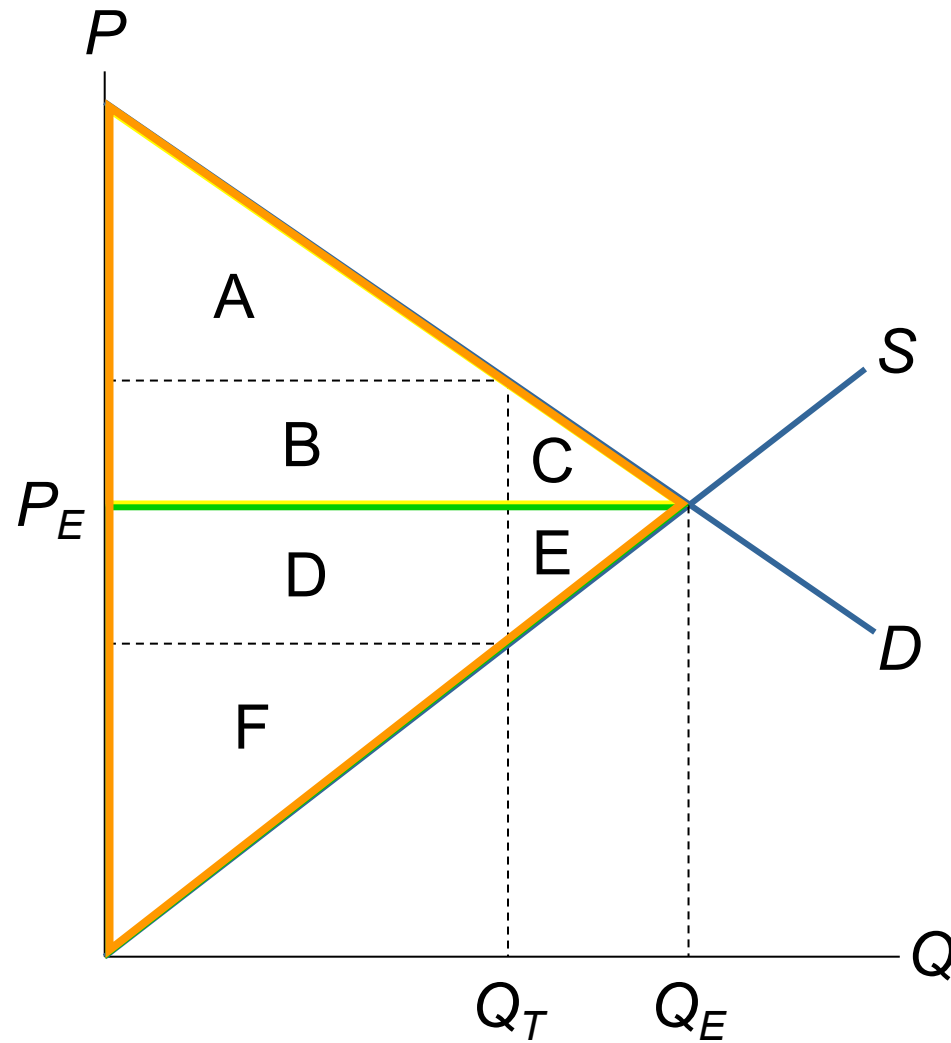
The Effects of a Price Ceiling

Without Price Ceiling,

$$CS = A + B + C$$

$$PS = D + E + F$$

$$\begin{aligned}\text{Total surplus} &= CS + PS \\ &= A + B + C \\ &\quad + D + E + F\end{aligned}$$



The Effects of a Price Ceiling

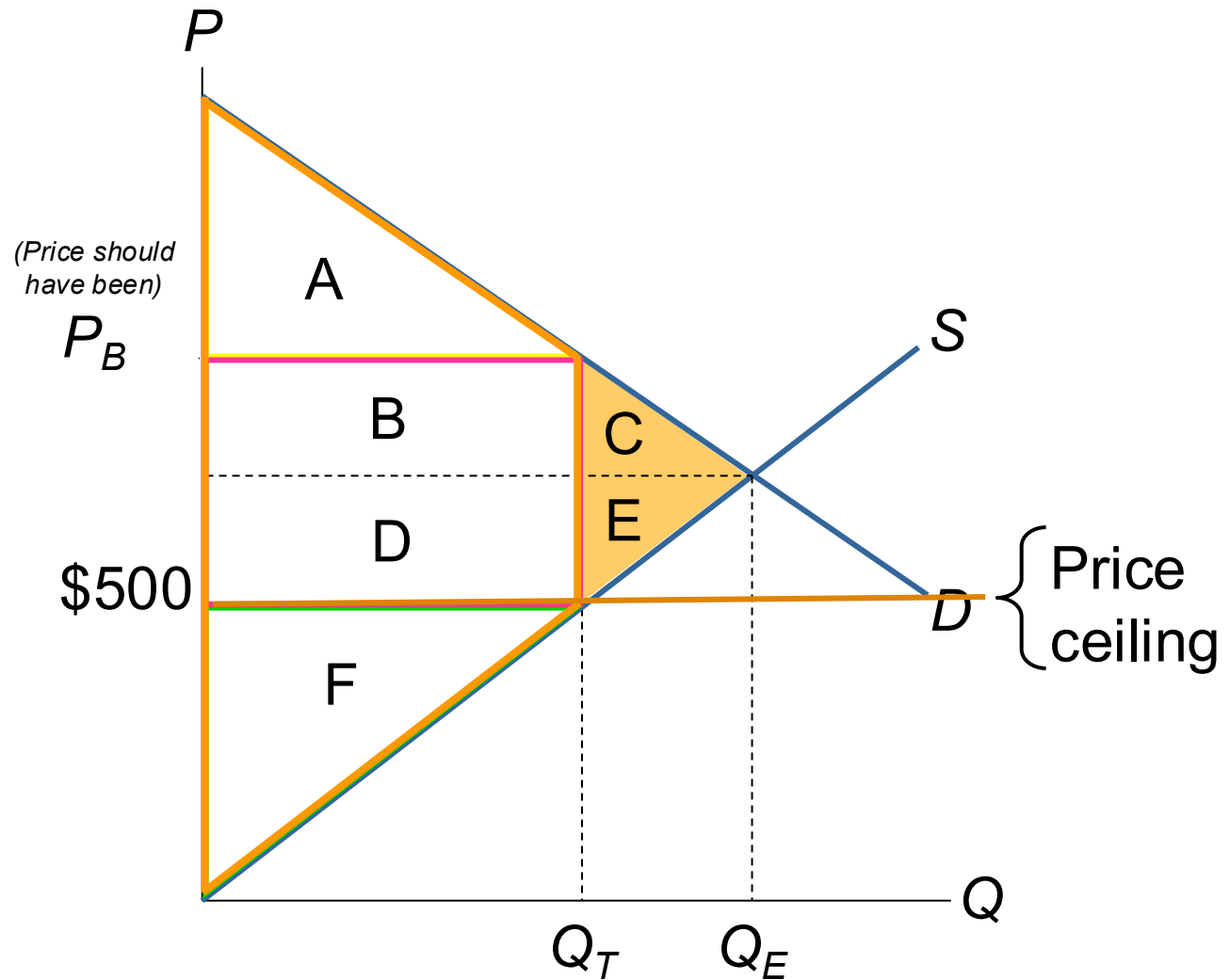
With Price Ceiling,

$$CS = A + B + D$$

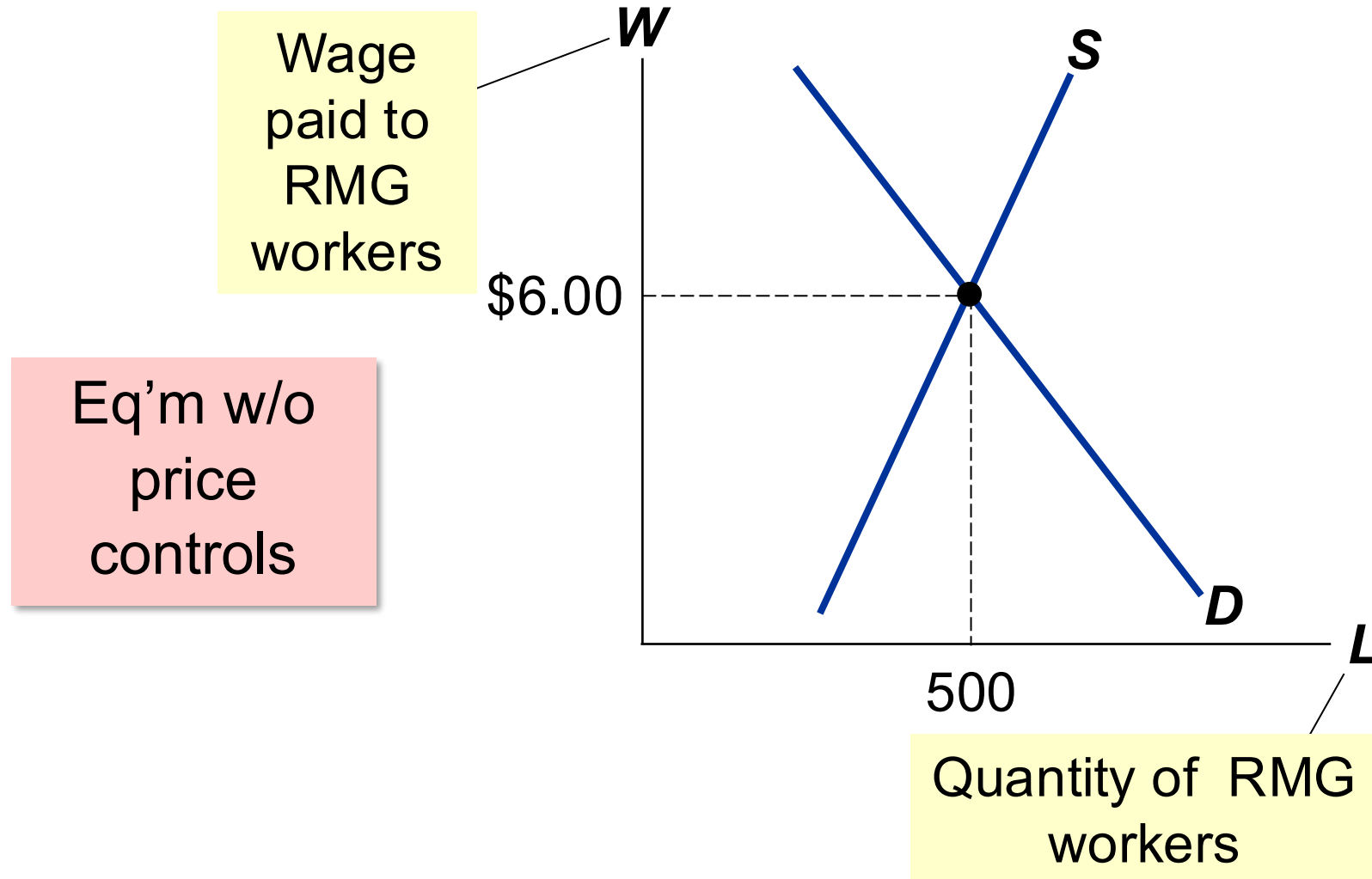
$$PS = F$$

$$DWL = C + E$$

$$\begin{aligned}\text{Total surplus} \\ &= A + B \\ &\quad + D + F + C + E\end{aligned}$$

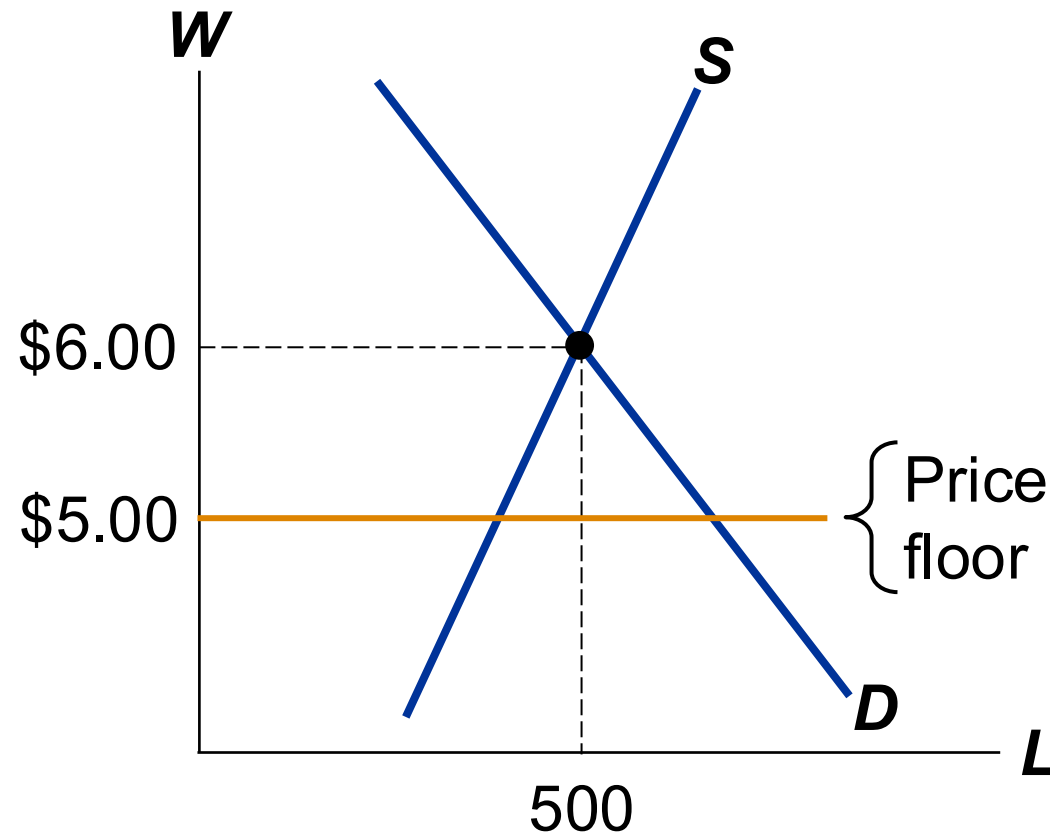


EXAMPLE 2: The Market for RMG Labor



How Price Floors Affect Market Outcomes

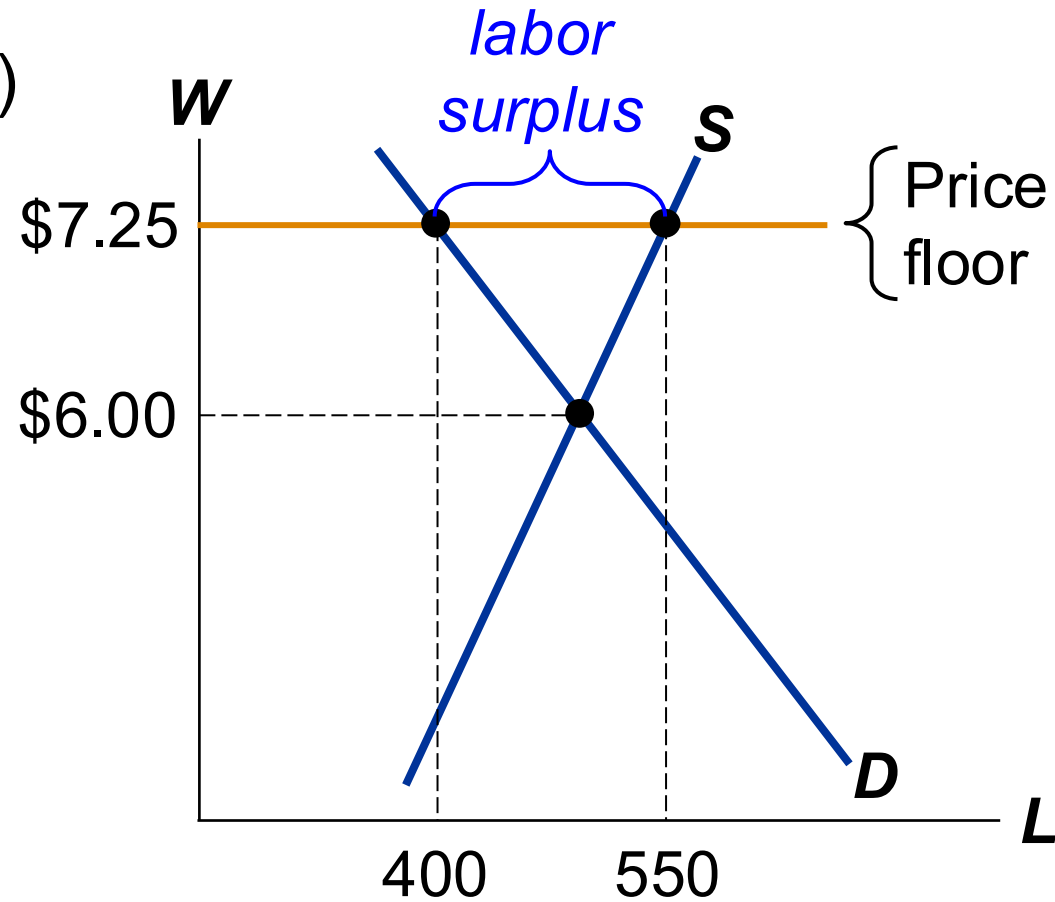
A price floor below the eq'm price is **not binding** – has no effect on the market outcome.



How Price Floors Affect Market Outcomes

The eq'm wage (\$6)
is below the floor
and therefore
illegal.

The floor
is a **binding
constraint**
on the wage,
causes a
surplus (i.e.,
unemployment).



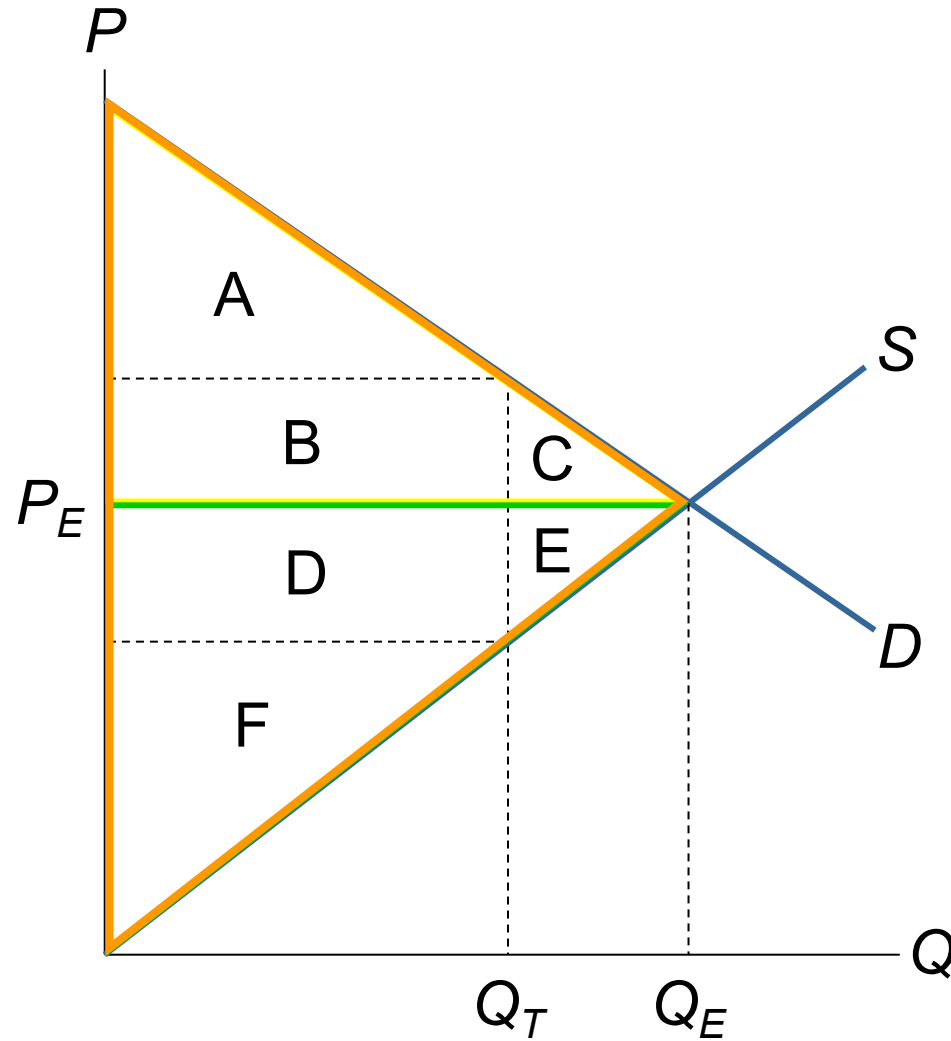
The Effects of a Price Floor

Without Price Floor,

$$CS = A + B + C$$

$$PS = D + E + F$$

$$\begin{aligned}\text{Total surplus} &= CS + PS \\ &= A + B + C \\ &\quad + D + E + F\end{aligned}$$



The Effects of a Price Floor

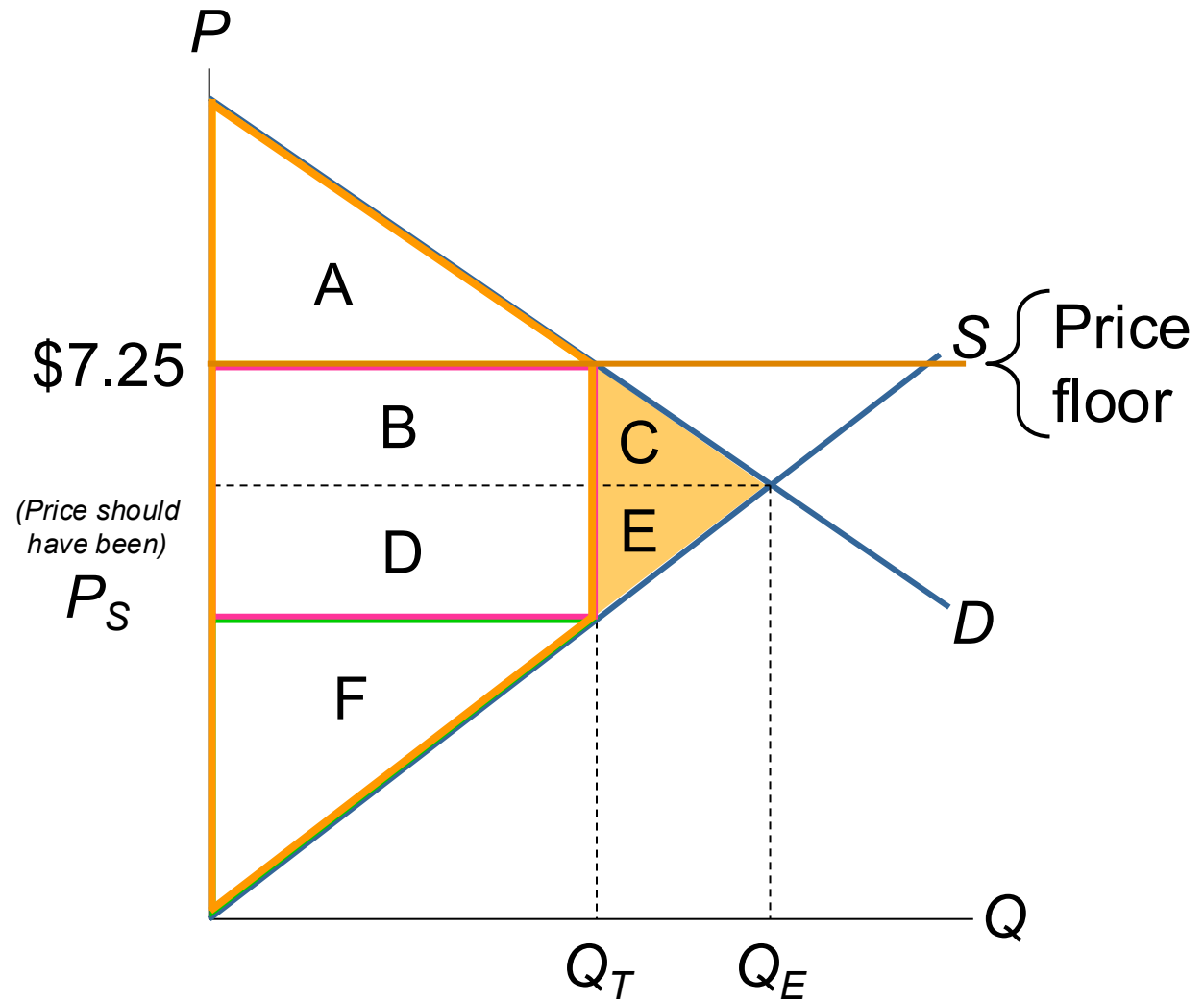
With Price Floor,

$$CS = A$$

$$PS = F + B + D$$

$$DWL = C + E$$

$$\begin{aligned} \text{Total surplus} \\ &= A + B \\ &\quad + D + F + C + E \end{aligned}$$



Summary

- The height of the D curve reflects the value of the good to buyers—their willingness to pay for it.
- Consumer surplus** is the difference between what buyers are willing to pay for a good and what they actually pay.
- On the graph, consumer surplus is the area between P and the D curve.

Summary

- The height of the S curve is sellers' cost of producing the good. Sellers are willing to sell if the price they get is at least as high as their cost.
- Producer surplus** is the difference between what sellers receive for a good and their cost of producing it.
- On the graph, producer surplus is the area between P and the S curve.

Summary

A **price ceiling** is a legal maximum on the price of a good.

A **price floor** is a legal minimum on the price of a good.